Test Coverage Report — Sprint 8

Final deliverable 6 — SENG 34213 System Development Project

Hotel Website (HW), Forever Hotel. Snapshot: 2026-10-25 (Asia/Colombo).

Assessment status: Unit/module coverage verified; complete release coverage not established

Source revision: `63673ba704b4ae645a8f067e246309b6fb0f03d0`. Branch: `docs/190-release-setup-and-demo-evidence`.

The reports were prepared after this commit; they do not certify a deployed release.

Scope is the HW repository plus identified API Gateway evidence, not the complete multi-subsystem hotel system.

Guideline source: SENG34213_System_Development_Project (1).pdf, section 10.1, printed page 40. The guideline suggests nested testing/security paths; these files are directly under `documents/` as requested by the user.

1. Purpose and criteria

Guideline sections 6.2 and 8.2 require at least 80% coverage on new code. Repository standards additionally require 90% branch coverage for critical business logic. Whole-tree coverage is reported here; a changed-line/new-code coverage calculation has not been performed. A global pass does not prove every new module or endpoint meets its target.

2. Execution summary

Scope	Tests passed	Lines	Branches	Functions	Statements	Boundary
Backend Jest unit suite	431/431	88.05%	82.16%	87.5%	88.02%	Co-located src unit tests; not all SQL paths.
Frontend selected TS modules	50/50	99.75%	98.49%	100%	Not emitted	Selected modules only; excludes TSX/rendered UI, smoke/browser suites.
API Gateway compiled suite	20/20	98.44%	90.36%	95.10%	Not emitted	All-files total includes compiled tests; see source-only rows below.
Load tooling: run.mjs	8/8	94.34%	90.30%	100%	Not emitted	Generator code; not measured capacity of deployed application.
Load tooling: metrics.mjs	Same 8 scenarios	100%	100%	100%	Not emitted	Metric calculation module.
Staging release-core	16/16	100%	86.96%	100%	Not emitted	Mock adapter; host SSH/Docker CLI not instrumented.
Staging smoke module	Same 16 scenarios	100%	96.36%	100%	Not emitted	Synthetic HTTP responses; not a deployed staging run.

Backend measured 1629/1850 lines and 1101/1340 branches across 49 unit suites. Its configured 80% global threshold passed. Do not average these percentages across different instruments/denominators.

3. Critical modules and uncovered boundaries

Backend module	Lines	Branches	Assessment
bookings/bookings.service.ts	95.65%	87.5%	Below 90%; gap remains
bookings/booking-management.service.ts	96.37%	84.54%	Below 90%; gap remains
bookings/quote-pricing.ts	95%	95.83%	Meets 90% branch target in this run
payments/payhere-payments.service.ts	100%	91.48%	Meets 90% branch target in this run
payments/payhere-provider.ts	100%	96.22%	Meets 90% branch target in this run
database/repositories/booking.repository.ts	20%	30%	Below 90%; gap remains
payments/payhere.repository.ts	5.63%	0%	Below 90%; gap remains

Booking orchestration and management remain below the repository's 90% critical-branch target. SQL repositories are primarily exercised by DB integration and have low unit coverage; this run cannot substitute unit coverage for SQL assertions.

4. Gateway source-only coverage

- app.js | 96.56 | 84.00 | 80.95 | 130-131 194-195 216-217 253-255
- config.js | 95.45 | 79.71 | 100.00 | 74-80
- cookies.js | 100.00 | 100.00 | 100.00 |
- errors.js | 100.00 | 100.00 | 100.00 |
- http.js | 100.00 | 100.00 | 100.00 |

Gateway config branch coverage is 79.71%; the all-files total also counts test code. No claim is made that every source file passes an 80% branch gate.

5. Integration, UI and exclusions

The fresh HTTP/security integration attempt reported 6/6 failed scenarios because database setup timed out before scenario assertions ran. Classify these as BLOCKED BY ENVIRONMENT, not a passing integration run or demonstrated business-logic defect. Historical fixture passes remain dated evidence only. Browser E2E, quality screenshots, real provider verification, pooled SQL race and staging deployment were not rerun for these reports.

Backend Jest excludes main.ts, spec files and interfaces; module definitions/repositories remain in its denominator. Frontend coverage selects imported .ts modules, excluding TSX, CSS, server actions and browser-rendered interactions. Staging CLI release.mjs and actual remote operations are not covered by the core/smoke figures.

6. Reproduction

- Backend: `node node_modules/jest/bin/jest.js --runInBand --coverage` from backend.
- Frontend: `npm run test:coverage` from frontend after its documented prerequisites.

- Gateway: `npm run test:cov` from the separate api-gateway checkout.
- Performance/staging tooling: `npm run test:coverage` in performance and deploy/staging.
- DB contracts: `npm run test:contracts` against a reachable disposable schema.

7. Required follow-up

Restore the disposable DB and rerun API/SQL tests; improve critical-branch cases; calculate release/new-code coverage explicitly; attach the reviewed CI and browser reports for the exact release SHA. These are open acceptance items, not completed work.

Evidence and sign-off

Current requirement/screen mapping: release traceability.

Prepared from executable results and source inspection; reviewer/supervisor acceptance is not supplied.

Do not attach real environment files, passwords, JWTs, card data or raw provider responses.